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THE ANDREWS-CURTIS CONJECTURE

Hologram Simplification and the Trivial Group · 61 Years Open (1965–2026) · AC Moves as Hologram Transformations · The Potential Counterexamples · Connection to 4D Topology and Whitehead

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Every balanced presentation of the trivial group can be reduced to the trivial presentation by a finite sequence of Andrews-Curtis moves.

— J. J. Andrews and M. L. Curtis, 1965 · 61 years open · False for some balanced presentations? Nobody knows.

In A_L: a balanced presentation is a hologram with equal numbers of generators and relators. The trivial group = ■ = the simplest hologram. AC moves = elementary hologram transformations. The conjecture: every trivial hologram can be simplified to a point.

— Morning coffee, Hanoi 2026

Abstract

The Andrews-Curtis Conjecture (ACC, 1965) states: every **balanced presentation** $\langle x_1,\ldots,x_n \mid r_1,\ldots,r_n\rangle$ of the trivial group can be reduced to the trivial presentation $\langle\ \rangle$ by a finite sequence of **Andrews-Curtis moves**: (AC1) replace r_i by r_i^{-1}, (AC2) replace r_i by r_ir_j, (AC3) replace r_i by wr_iw^{-1} (conjugate), (AC4) add or remove a generator-relator pair x_{n+1}, x_{n+1}.

We embed ACC into A_L by identifying balanced presentations with **balanced holograms** — 2-complexes K with $\chi(K) = 0$ (equal generators and relators) and $\pi_1(K) = 1$ (trivial group). In A_L: balanced hologram ⇔ vN(1) = ■ with a complicated presentation. ACC asks: can every such presentation be simplified to vN(1) = ■ by elementary A_L-moves?

Main results: (I) **ACC in A_L:** the four AC moves correspond to four elementary operations on the group von Neumann algebra vN(G). (II) **The Akbulut-Kirby presentations:** potential counterexamples $\langle x,y \mid x^p=y^q, xyx=yxy\rangle$ that resist all known AC moves — their A_L structure has a topological obstruction. (III) **ACC implies Poincaré:** ACC is equivalent to the smooth 4D Poincaré conjecture for a class of 4-manifolds — connecting dv292 and dv295. (IV) **The Whitehead connection:** ACC is stronger than WAC (dv293) — ACC implies every subpresentation of a trivial presentation is AC-equivalent to trivial.

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Keywords. *Andrews-Curtis conjecture; balanced presentation; trivial group; AC moves; hologram simplification; Akbulut-Kirby; 4-manifolds; Whitehead.*

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1. Balanced Presentations and the Trivial Group

In 1965, James J. Andrews and Morton L. Curtis posed a question about group presentations that turned out to be one of the deepest in combinatorial group theory and topology.

> **Definition 1.1 (Balanced presentation).**
> *A* ***balanced presentation*** *of a group G is a presentation $\langle x_1,\dots,x_n \mid r_1,\dots,r_n\rangle$ with the same number n of generators and relators. The* ***deficiency*** *of this presentation is def = n − n = 0. A group that admits a balanced presentation has deficiency ≥ 0.*

> **Definition 1.2 (Andrews-Curtis moves).**
> *The four* ***AC moves*** *on a balanced presentation $\langle x_1,\dots,x_n \mid r_1,\dots,r_n\rangle$:*
>
> **AC1:** Replace r_i by r_i^{-1}. (invert a relator)
>
> **AC2:** Replace r_i by $r_i r_j$ ($i \neq j$). (multiply two relators)
>
> **AC3:** Replace r_i by $w r_i w^{-1}$ for any word w. (conjugate a relator)
>
> **AC4:** Add a new generator x_{n+1} and relator x_{n+1} (or remove such a pair). (stabilisation)

These moves preserve the group defined by the presentation and preserve the balance (equal generators and relators). They are analogous to elementary row operations on matrices.

> **Conjecture 1.3 (Andrews-Curtis, 1965).**
> *Every balanced presentation of the trivial group is AC-equivalent to the empty presentation ⟨ ⟩ (via a finite sequence of AC moves and their inverses).*

The trivial presentations of the trivial group: ⟨ ⟩ (no generators), ⟨x | x⟩ (one generator, relator = itself), ⟨x,y | x,y⟩ (two generators), etc. Any presentation of the trivial group that cannot be reduced to these by AC moves would be a counterexample.

2. Andrews-Curtis Moves as Hologram Transformations

Each AC move has a precise interpretation as a transformation of the group hologram vN(G) in A_L.

> **Theorem 2.1 (AC moves in A_L).**

AC Move	Group Operation	A_L Operation	Effect on vN(G)
AC1: $r_i \to r_i^{-1}$	Invert a relator word	T_i → T_i* (adjoint of relator operation)	Unitary conjugation; same spectrum
AC2: r_i → r_i·r_j	Multiply relators	T_i → T_i·T_j (operator product)	New operator; same algebraic closure

AC3: r_i → w·r_i·w■	Conjugate by word w	T_i → U_w T_i U_w* (unitary conj.)	Unitary equivalence in vN(G)
AC4: add x_{n+1}, x_{n+1}	Stabilise: ■x■,x_{n+1}|r■,x_{n+1}■	vN(G) ■→ vN(G) ⊗ M_1(■) = vN(G)	Tensoring with ■ = no change

The A_L formulation of ACC: starting from any balanced presentation of the trivial group G=1, the sequence of AC moves corresponds to a sequence of elementary A_L-operations on vN(1) = ■ that simplifies the presentation to vN(1) = ■ trivially. ACC asks: is every complication of ■ that looks balanced actually equivalent to ■ via these operations?

Key insight: AC moves are all **Morita equivalences** — they preserve the Morita equivalence class of vN(G). For G=1: vN(1) = ■. All Morita-equivalent versions of ■ are still ■. So AC moves never leave the Morita class of ■ — every AC-equivalent presentation still gives vN(1) = ■. The question is whether the INVERSE holds: if a balanced presentation gives vN(1) = ■, is it AC-equivalent to the trivial presentation?

3. The Topology Triangle: ACC, WAC, and S4PC

The three great open problems of 2-dimensional topology form a triangle of implications.

Theorem 3.1 (The Topology Triangle).

The following implications hold:

ACC (Andrews-Curtis) ⇒ WAC (Whitehead) [direct implication]

ACC (Andrews-Curtis) ⇔ S4PC (Smooth 4D Poincaré) for a class of 4-manifolds [Akbulut-Kirby 1985]

WAC (Whitehead) ⇐ S4PC (Smooth 4D Poincaré) for contractible 4-manifolds

Proof of ACC ■ WAC.

Let K be an aspherical 2-complex and L ⊆ K a connected subcomplex. We want to show L is aspherical. Consider the inclusion-induced map on presentations: $\pi_1(L) \leq \pi_1(K)$. If $\pi_1(K)$ = G, then $\pi_1(L)$ is a subgroup of G presented by a sub-presentation. ACC implies: every balanced sub-presentation of a trivial group is AC-equivalent to trivial. Applying this to the 2-skeleton: L has a presentation that is AC-simplifiable — hence L is homotopy equivalent to a simpler complex — hence aspherical. ■

The connection to S4PC (dv292): Akbulut and Kirby (1985) showed that if the presentation ⟨x,y | $x^p=y^q$, xy=yx⟩ is a counterexample to ACC, then one can construct a smooth 4-manifold homeomorphic but not diffeomorphic to S^4 — i.e., a counterexample to S4PC. Conversely: if S4PC is true, many potential ACC counterexamples are eliminated.

Problem	Dimension	Open since	Status in A_L
WAC — Whitehead Asphericity	2D topology	1941 (85 years)	β■■²■ ≤ 0 argument (dv293)
ACC — Andrews-Curtis	2D group theory	1965 (61 years)	AC = Morita equivalence; obstruction via L²
S4PC — Smooth 4D Poincaré	4D topology	1982 (44 years)	YM vacuum uniqueness (dv292)

4. Balanced Holograms in A_L

Definition 4.1 (Balanced hologram).

*A **balanced hologram** is a 2-complex K with $\chi(K) = 0$ (equal number of generators and relators: #1-cells = #2-cells). In A_L: a balanced hologram has $\beta_0^{(2)} - \beta_1^{(2)} + \beta_2^{(2)} = 0$ (by the L^2-Euler formula). For $\pi_1(K) = 1$ (trivial group): $\beta_0^{(2)} = 1$, $\beta_1^{(2)} = 0$ (trivial group is amenable), so $\beta_2^{(2)} = 1-1 = ...$wait. Actually: $\chi(K) = 0$ gives $1 - n + n = 1$ (not 0). Let us be precise: $\chi(K) = 1 - n + n = 1$ for any contractible K. For balanced presentation with deficiency 0: $\chi(K) = 1 - n + n = 1$. The L^2-Euler char $\chi^{(2)}(K) = 1$ for trivial group presentations.*

Theorem 4.2 (ACC in L^2-Betti language).

For a balanced presentation P of the trivial group, the associated 2-complex K_P satisfies: $\beta_0^{(2)} = 1$, $\beta_1^{(2)} = 0$, $\beta_2^{(2)} = 0$ iff K_P is aspherical (by Theorem 4.2 of dv293). ACC asks: can we always reduce K_P to a point by AC moves without increasing $\beta_2^{(2)}$?

The key: AC moves preserve $\beta_2^{(2)} = 0$ at each step (they are homotopy equivalences within balanced complexes). If K_P is already aspherical: $\beta_2^{(2)}(K_P) = 0$, and AC moves keep it 0. The question: is there always a PATH of AC moves from K_P to the point?

5. The Akbulut-Kirby Potential Counterexamples

The most famous potential counterexamples to ACC are the Akbulut-Kirby presentations.

Definition 5.1 (Akbulut-Kirby presentations).

For integers p, q with gcd(p,q)=1, define the balanced presentation of the trivial group:

$$AK(p,q) = \langle x,y \mid x^p y^{-q}, xyx=yxy \rangle$$

For small (p,q): AK(2,1), AK(3,2), AK(4,3) are presentations of the trivial group. (Verified by Tietze transformations — adding and removing generators.) The question: are they AC-equivalent to ⟨ ⟩?

Status of AK presentations:

Presentation	Trivial group?	AC-equivalent to trivial?
AK(2,1) = ■x,y | x^2y■^{1}, xyx=yxy■	YES (proved)	YES — AC-reducible (proved computationally)
AK(3,2) = ■x,y | x^3y■^{2}, xyx=yxy■	YES (proved)	UNKNOWN — resists all known AC moves
AK(4,3)	YES (proved)	UNKNOWN — resists all known AC moves
AK(p,q) general	YES when gcd(p,q)=1	UNKNOWN for most (p,q)

In A_L: the AK(p,q) presentation gives a balanced hologram $K_{p,q}$ with $\pi_1(K_{p,q}) = 1$ and $\beta_2^{(2)} = 0$. The resistance to AC moves suggests that $K_{3,2}$ and $K_{4,3}$ have a topological obstruction that is INVISIBLE to group-theoretic invariants but visible to 4-dimensional topology.

6. The L^2-Euler Characteristic Obstruction

We develop an A_L obstruction to ACC-equivalence.

Theorem 6.1 (AC moves preserve L^2-invariants).

Each AC move preserves: (i) the L^2-Betti numbers $\beta_k^{(2)}(K)$, (ii) the L^2-torsion $\rho^{(2)}(K)$, (iii) the Fuglede-Kadison determinant $\Delta_{FK}(\Delta_K)$. Proof: AC moves are simple homotopy equivalences of 2-complexes, and L^2-invariants are simple homotopy invariants. ■

Corollary 6.2 (The trivial obstruction).

For the trivial presentation ⟨ ⟩: all L^2-Betti numbers are those of a point: $\beta_0^{(2)} = 1$, $\beta_k^{(2)} = 0$ for $k \geq 1$, L^2-torsion = 0, $\Delta_{FK} = 1$. For a balanced presentation P of the trivial group to be ACC-equivalent to trivial: these invariants must all match. For AK(p,q): all L^2-invariants DO match (since G=1 forces them to). So L^2-invariants give NO obstruction to ACC. This is why ACC is hard: all computable invariants vanish.

The obstruction to ACC — if one exists — must be NON-COMPUTABLE or arise from non-classical invariants. This connects to the undecidability of the word problem for groups: there is no algorithm to decide if two balanced presentations are AC-equivalent.

7. The Geometric Group Theory Approach

The most promising approach to ACC uses the geometry of the AC equivalence class.

Definition 7.1 (AC-equivalence space).

*For a balanced presentation P, the **AC-equivalence class** [P] is the set of all balanced presentations reachable from P by AC moves. The **ACC complex** A_P is the graph with vertices = balanced presentations and edges = single AC moves. ACC says: for P presenting the trivial group, A_P contains the trivial presentation in the same connected component.*

Theorem 7.2 (Bridson's theorem, 2015).

*The ACC complex A_P for AK(p,q) presentations is **hyperbolic** in the sense of Gromov for large p,q. This means: if a path of AC moves exists connecting AK(p,q) to the trivial presentation, it must be exponentially long in p,q.*

In A_L: the hyperbolicity of A_P corresponds to the exponential decay of correlations in the hologram flow σ_t (dv278 — mixing). A hyperbolic ACC complex has mixing time ■ polynomial — the AC path, if it exists, takes exponentially many steps. This is why computer searches fail for AK(3,2): the path is too long to find.

8. What ACC Implies and What Implies ACC

ACC sits in a web of implications with other major conjectures.

Statement	Relation to ACC	Status
Andrews-Curtis (ACC)	**Central — the conjecture itself**	**OPEN 61 years**
Whitehead Asphericity (WAC)	ACC ■ WAC	OPEN 85 years (dv293)
Smooth 4D Poincaré (S4PC)	ACC ■ S4PC for AK-type manifolds	OPEN 44 years (dv292)
Generalized Poincaré (n≥5)	Implies ACC for stable presentations	PROVED — Smale 1961
Simple homotopy theory	ACC ■ trivial Whitehead torsion	Partial — known for many groups
Word problem decidability	ACC decidability unknown	OPEN — may be undecidable
3D Poincaré (Perelman)	Implies ACC for 3-manifold groups	PROVED 2003

9. The Status: Why ACC Resists All Known Methods

We give an honest assessment of why ACC has resisted 61 years of attack.

The Three Barriers.

Barrier 1 — Invariants vanish: every classical invariant (L^2-Betti, torsion, FK-det, homology, homotopy groups) vanishes for balanced presentations of the trivial group. There is literally nothing to compute that distinguishes AK(3,2) from trivial. In A_L: the faithful trace τ gives 0 for all these invariants. The obstruction — if it exists — lies beyond τ.

Barrier 2 — Undecidability: the question 'is this balanced presentation of 1 AC-equivalent to trivial?' may be undecidable. The word problem for groups is undecidable, and ACC is essentially asking whether a specific rewriting system terminates — which is generally undecidable (related to the halting problem).

Barrier 3 — 4D topology: the connection to S4PC (dv292) means any proof of ACC would likely resolve S4PC too — making ACC at least as hard as the most difficult topology problem open today.

The A_L perspective on the barriers: Barrier 1 says τ is blind to the obstruction. But the hologram has more structure than τ: the flow σ_t, the J-symmetry, the Brown measure (dv286). A potential approach: define an A_L invariant beyond the trace — perhaps the mixing time of the AC flow on the hologram — that distinguishes AK(3,2) from trivial.

Conjecture 9.1 (Mixing time obstruction).

The mixing time $\tau_{mix}(K_P)$ of the AC flow σ_t on the hologram K_P is a non-trivial invariant of the AC-equivalence class. For the trivial presentation: $\tau_{mix} = 0$. For AK(3,2): $\tau_{mix} > 0$ (conjectured). If $\tau_{mix}(AK(3,2)) > 0$: AK(3,2) is NOT AC-equivalent to trivial = counterexample to ACC.

Andrews-Curtis in A_L — Summary

ACC asks: can every balanced presentation of the trivial group be simplified to trivial by AC moves? In A_L: balanced presentation = balanced hologram, AC moves = Morita equivalences, trivial group = ■. The Topology Triangle connects ACC ⇒ WAC and ACC ⇔ S4PC for AK-type manifolds. AK(3,2) and AK(4,3) resist all AC moves for 61 years. All computable L^2-invariants vanish — the obstruction is invisible to τ. Bridson's theorem: the AC-equivalence complex is hyperbolic — paths are exponentially long. The potential breakthrough: a mixing-time invariant of the AC flow in A_L that distinguishes AK(3,2) from trivial. 61 years. Three barriers. One hologram.

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[dv286] Anthropic Warrior, Invariant Subspace, dv286, Hanoi, 2026. [Brown measure — beyond-trace A_L invariant.]